Benjamin Greenberg

Hardware & Embedded Systems Engineering
Swarthmore, PA | bgreenb2@swarthmore.edu | linkedin.com/in/benjamin-greenberg- | github.com/BenGreenberg07

EDUCATION

Swarthmore College | Swarthmore, PA May 2029
Candidate for B.S. in Engineering and B.A. in Economics | GPA: 3.93/4.00 | Sigma Xi

ENGINEERING EXPERIENCE

Haptic Robotics Research Assistant | Swarthmore College October 2025 – Present
- Configuring a PHANToM Premium device for post stroke rehabilitation research in a neuromuscular hand control lab
- Created and 3D-printed a custom attachment enabling specific finger interaction instead of whole hand gripping
- Built C++ force feedback and position tracking software on the OpenHaptics SDK, extending its limited functionality
- Designing a patient testing protocol to assess whether post stroke patients show a mix of sensorimotor deficits rather than one dominant type, informing future rehabilitation approaches

Computational Network Dynamics Researcher | Lehigh University June 2025 – Present
- Independently built and ran an agent based epidemiological model (4,000 agents, 50 replicates), showing that small dips in vaccination coverage can trigger sudden, hard to reverse collapses in herd immunity
- Authored a first-author manuscript submitted to the Q1 Journal of the Royal Society Interface, currently under review

Aerial Robotics Researcher | Lehigh University May 2026 – August 2026
- Engineered an autonomous blimp drone for rehabilitation applications in the Aerial Swarms Laboratory
- Designed and 3D-printed a custom drone frame integrating an ESP32 microcontroller and IMU, achieving a 29g total system weight to minimize the motor power needed for lift, iterating the frame design over 10+ builds
- Wrote a pure-pursuit & PD autonomous controller in C, reducing post-push position error 67% (1.27 to 0.42 m) and returning the blimp to within 1.5 m of target in 95% of trials with a 13.2 s median recovery, under motion capture
- Authoring a second-author manuscript on human-drone interaction and intent recognition, submitting to IEEE RA-L

PROJECTS

SEPTA Display May 2026 – Present
- Iterating a 16-panel HUB75 LED departure board (Raspberry Pi 5, 5V/100A) left as a rough senior design prototype
- Rebuilt the Python display layer with live train position from SEPTA TrainView GPS. Redesigning the enclosure for weatherproof mounting and coordinating station installation with SEPTA

Cantilever Truss | ENGR 006 April 2026
- Designed, machined, and load-tested a 6.45 lb aluminum cantilever truss that carried a 1,000 lb point load

Macropad May 2026
- Designed schematic and PCB in KiCad for a 9-key macropad, manually soldered it, and flashed custom QMK firmware

MakerWorld December 2024
- Published 10 3D-printable models in Fusion 360, drawing 5M+ views and 22K+ downloads to date

SKILLS & COURSEWORK

Tools & Lab: Soldering, continuity testing, FDM 3D printing, tensile testing, Raspberry Pi, ESP32 integration
Software & CAD: C, C++, Python, MATLAB, KiCad, SolidWorks, Autodesk Fusion, ROS; Adobe, Google, & Microsoft Suite
Coursework: Circuit Analysis, Differential Equations, Engineering Dynamics (in progress, Fall 2026); Linear Systems Analysis (expected, Spring 2027); Electricity and Magnetism, Engineering Mechanics, Multivariable Calculus (Completed)
Languages: English (fluent), French (fluent)

LEADERSHIP EXPERIENCE

Swarthmore Robotics Club, *President* February 2026 – Present
- Leading a team developing an F1TENTH autonomous racing car, integrating LiDAR-based SLAM and ROS navigation

Swarthmore College Admissions, *Campus Tour Guide* November 2025 – Present
- Guiding 60 minute campus tours for potential students, selected for leadership, communication and interpersonal skill

Video & Graphic Design Business, *Founder* September 2020 – Present
- Founded and grew an online Minecraft art business to 55,000+ YouTube subscribers and 86,000+ TikTok followers

Swarthmore Clubs: Club Squash Team (Captain), Ceramics Club (Founder), Chabad (Executive Board Member)